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TOWARDS A MEANINGFUL CONCEPT OF POLLUTION IN INTERNATIONAL LAW

By

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THE question has been posed by, among others, C. B. Bourne: " What are you talking about when you are talking about pollution? What is pollution? How would you define it if you are going to remove the concept of damage from it? "[1] The fact is that " pollution " is a word whose precise meaning in law, particularly international law, is not easily discerned. It has been used in a wide variety of contexts, from international conventions to pessimistic speeches about the state of the environment, to describe different levels and kinds of man-induced[2] changes in the natural world. Within a particular context, pollution assumes a meaning, either explicitly stated or implicitly developed, which may bear little resemblance to its usage elsewhere. While any definition may appear satisfactory for the purpose to which it is applied, this proliferation of usages has led some publicists to believe that the term has lost any legal value it might once have had and should be discarded. Indeed, in many cases, pollution has been replaced by other words, supposedly more precise. Yet " pollution " is an extremely important concept in the contemporary discussion of international environmental law and an integral part of many international agreements, including the Stockholm Declaration on the Human Environment of 1972.[3] The problem of what constitutes a legally significant effect on the environment cannot be solved by the simple manipulation of labels.

By setting forth the ways " pollution " has been used in an international legal context in the recent past, this paper attempts to reveal the varying approaches taken which now make it difficult to consider " pollution " a reasonably precise and generally accepted concept in international law. The classification of these approaches could

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[1] C. B. Bourne, " Trail Smelter Arbitration: Oral Proceedings " (1971) 50 *Oregon Law Review* 291.

[2] If there is any clear consensus on any aspect of what pollution is, it is the general belief that pollution in the legal sense is necessarily caused by man, either directly or indirectly. This will be assumed in the discussion that follows.

[3] *Stockholm Declaration on the Human Environment*, Report of the United Nations Stockholm Conference on the Human Environment, Stockholm, June 5-16, 1972, A/Conf. 48/14.

be based on an evolution of the meaning of the term through time, or on the different attitudes taken toward environmental change in particular parts of the biosphere (atmosphere, oceans, and land), or on the degree of control and interest those using the label have had in the exploitation of the environment to be protected. Each of these factors is clearly related to the way the concept is used in a specific case and must be kept in mind in understanding its meaning in that context. Yet pollution, to be a useful concept, should have or develop a meaning which transcends and can be detached from particular usages.

An initial step must be taken to clarify the nature of the concept to be discussed. The word "pollution" has been used in two distinct senses: first, to indicate any alteration in a given environment, and, secondly, to indicate a threshold level of damage or interference which is legally significant. This paper adopts the second approach. What is important is the point at which the label is attached and the criteria deemed necessary for that attachment, not the specific label employed. Thus, some licence has been taken in incorporating into this analysis examples which appear to approach essentially the same threshold under the guise of another label. This study is limited to a description of what is meant by this threshold as an independent concept. The legal consequences which flow from the attachment of the label—questions of degrees of liability, and so forth—are important, but constitute fields of investigation which are beyond the scope of this paper. It is in the latter area that most of the discussion of international pollution to date has taken place.[4]

The paper is divided into two parts. In the first are outlined what the author considers the basic conceptual categories within which pollution has been approached in the past. The primary concern is the criteria required to pass the threshold of legal significance. The type of criteria employed is the basis on which these categories are distinguished, while the threshold level of "necessary" criteria and the area of application are used to show the range of definitions within each category. The second section of the paper discusses the two basic "reactions" to the limitations of these past approaches: wide definitions of pollution encompassing two or more types of

[4] This is not to deny the interrelation in specific cases of the threshold of label attachment and the attendant legal consequences. As Charles R. Ross shows with the 1909 Boundary Waters Treaty between the United States and Canada, a State may be willing to accept a "low" threshold of legally-significant environmental change only on the condition that the consequences of the label attachment be relatively lenient or only applied in very serious cases. Charles R. Ross, "National Sovereignty in International Environmental Decisions" (1972) 2 *Natural Resources Journal* 243. Yet it is this interrelationship which has been responsible for some of the confusion over what pollution means. To understand the meaning and significance of the label, one must attempt to set it apart from its legal consequences.

criteria, and new concepts to replace pollution. Finally, an attempt is made to place within this perspective present efforts to give the word " pollution " a generally acceptable meaning for the future, not limited by specific usages and concepts.

I. PAST APPROACHES TO POLLUTION

Five categories have been chosen to describe the range of approaches to pollution. The first two, " Pollution as any alteration of the existing environment " and " Pollution as the right of the territorial sovereign," are employed to set the extreme boundaries of the discussion and thus are presented more as ideal types than as categories of widely-shared viewpoints. The other three, " Pollution as damage," " Pollution as interference with other uses of the environment," and " Pollution as exceeding the assimilative capacity of the environment," correspond far more realistically to the major centres of opinion within the often unconscious controversy over the meaning of pollution.

A. *Pollution as any Alteration of Existing Environment*

In its purest form, this approach would allow little room for man in the natural world. By reference to it, a single drop of refined oil in the Pacific Ocean would become pollution under international law, as indeed would the most minimal change anywhere within man's reach. The distinction made earlier between pollution as any alteration and as a threshold of man-induced change breaks down here and, in this pure form, the approach is clearly meaningless as a general basis of law.

Yet in specific situations something similar to this has been adopted. A 1929 frontier agreement between Germany and Belgium contains the clause:

> Before erecting any building which might alter the present situation and have a *prejudicial* effect on the quality of the water, the Governments of both States shall come to an agreement with regard to the necessary protective measures. *In no case* is it permissible to let water *injurious to public health flow into* streams and ditches.[5]

Set forth is a very low threshold of acceptable alteration of a given body of water, mitigated only slightly by the vague reference to " prejudicial " effects. The second sentence appears, at first glance, to be setting more concrete standards for the degree of environmental effect, yet nowhere is there stated any threshold of change in the

[5] Agreement Concerning the Frontier Between Germany and Belgium, Aix-La-Chapelle, Nov. 7, 1929, League of Nations, Treaty Series (LNTS), Vol. 121, No. 795, Art. 60 (2), p. 361 (emphasis added).

river itself or any permissible degree of damage which its " polluted " waters might cause. What is said is that no amount of a particular type of water, described as " injurious to public health " may be added to streams and ditches.

A more recent example of the " purity " approach, in the highly specific area of radioactive waste, is provided by the Antarctic Treaty of 1959 which prohibits " any nuclear explosions in Antarctica and the disposal there of radioactive waste material." [6] Similarly the 1963 Treaty Banning Nuclear Weapon Tests in the Atmosphere, in Outer Space, and Under Water, " desiring to put an end to the contamination of man's environment by radioactive substances," [7] permits no nuclear explosions in the three parts of the biosphere mentioned in the title. Article I (1) (b) is particularly important in showing the environmental component of this absolute prohibition. The presence of any level of radioactive debris outside the territory of the testing State is considered unacceptable and, in itself, constitutes pollution in the sense of this paper.[8] This view of the need to keep as much of the planet free of radioactive waste as possible is further embodied in the General Assembly's Resolution on the Urgent Need for Suspension of Nuclear and Thermonuclear Tests. The Assembly's primary concern is stated to be the injury " to human and animal life " and the fear that " such fall-out may poison the Earth's atmosphere for decades to come." [9] Yet the threshold of pollution depends not upon a particular level of damage to man or the environment but upon the detonation of a nuclear weapon and the subsequent emission of radioactive fallout. Once again, a standard of absolute purity is called for.

The applicability of this approach is clearly limited in today's world to exactly the sort of specific and blatant forms of environmental degradation presented by nuclear explosions. Absolute purity can only be a useful concept when the interaction between the potential pollutant and a certain part of the biosphere would be a relatively rare, controllable, and avoidable occurrence. As long as radioactive wastes could be disposed of in the oceans and nuclear weapons tested underground, it was plausible for negotiators to accept the " absolute " standards represented in the documents discussed above. Yet, in addition to problems of political acceptability of such stan-

[6] Antarctic Treaty, Washington, Dec. 1, 1959, United Nations, Treaty Series (UNTS), Vol. 402, Art. v (1), p. 71.

[7] Treaty Banning Nuclear Weapon Tests in the Atmosphere, in Outer Space and Under Water, Moscow, Aug. 5, 1963, UNTS, Vol. 480, p. 45.

[8] *Ibid.*, Art. I (1) (b).

[9] Resolution on the Urgent Need for Suspension of Nuclear and Thermonuclear Tests, United Nations, General Assembly, Resolution No. 2828 (XXVI), Dec. 16, 1971.

dards for countries like France and China whose nuclear programmes required further atmospheric testing, the difficulty of adhering to the " purity " approach (at least as long as underground testing is permitted) was quickly revealed. An underground detonation by the USSR on January 15, 1965, in the Semipalatinsk area of Soviet Central Asia released radioactive debris detected by U.S. reconnaissance planes as having crossed Soviet borders in apparent violation of the Test-Ban Treaty's terms. However, unwilling to adopt a " strict " viewpoint which might later be applied against its own tests, the United States refrained from directly accusing the Soviet Union of a treaty violation, instead considering it an " inadvertent " and " isolated " incident. A similar stance was taken by the Soviet Union in 1966 when a U.S. explosion in Nevada vented radioactive fallout.[10] What evolved in this exchange was an understanding that something more than the detection of *any* radioactivity would be required to charge a signatory of the treaty with an illegal act. The " absolute " standards adopted earlier were, in fact, replaced by a threshold of *unacceptable* radioactivity, a threshold undefined either in terms of the amount which actually crossed an international border or the degree of the threat it posed to human or animal life.

In the case of the German-Belgian frontier treaty, another problem with the approach seems clear. To label a minimal influx of " injurious " waters " pollution " is to ignore the assimilative capacity of the stream and bestow legal significance on a situation that may not require it. Should this become a standard to maintain, many worthwhile developments along the affected waterway might well be impossible. If, instead, what the agreement seeks to prevent is an unhealthful body of water, the threshold of pollution should be defined either in terms of water quality standards for that body of water or by reference to far more precise emission standards which take into account each stream's assimilative potential.[11]

B. *Pollution as Right of Territorial Sovereign*

In contrast to the previous approach, this view in its ideal form allows no international purview over any type or degree of environmental

[10] This problem received a limited amount of publicity in several *New York Times* articles and an editorial of Jan. 22, 1965. John W. Finney, " U.S. Says Soviets May Have Broken Test-Ban Treaty," *New York Times*, Jan. 20, 1965, p. 1, col. 1; " U.S. Cautious on Fall-out from Soviet," *New York Times*, Jan. 21, 1965, p. 4, col. 4; " The Use of Nuclear Explosions," editorial, *New York Times*, Jan. 22, 1965, p. 42, col. 1; " A-Test Violation Denied by Soviet," *New York Times*, Jan. 26, 1965, p. 8, col. 3; " Soviet Test Blast is Assessed by U.S.," Mar. 10, 1966, p. 13, col. 1; " Nuclear Accident in Nevada is Assailed by Soviet Press," May 1, 1966, p. 71, col. 4.

[11] This distinction is an important one and will be explored at greater length below.

change which arises within the geographical borders of a sovereign State, regardless of the extra-territorial effects of the change. It does not speak directly to the question of pollution in areas outside of State control (the oceans, outer space, and unclaimed land), yet within the State it implies a very high or infinite threshold of alteration for which international law would be unable to hold the State responsible.

Such an extreme position finds little support in contemporary discussions of international pollution, but the " Harmon Doctrine " can be cited as within the conceptual framework of this approach. Responding to a Mexican protest over the United States' diversion of the waters of the Rio Grande, the United States Attorney-General, Judson Harmon, saw no principle of international law which would bar the United States from or make it liable for such a diversion. Comity might serve to limit such actions but " that question should be decided as one of policy only." [12] While this particular case involved the quantity of water, the implications as to quality are obvious. Those portions of rivers within United States jurisdiction could be used as the United States saw fit. International law set no limits of permissibility to the external effects such use might entail. Issued in 1895, the Harmon Doctrine reflected a far less interdependent international situation than that which would develop and be recognized in the twentieth century by the evolution of riparian rights theory.

In addition to the present unacceptability of the general viewpoint, thinking of pollution as the right of the territorial sovereign leads, as Lester points out, to the logical result that " unlimited exercise of sovereignty by an upper riparian State must limit the equally important sovereignty of the lower riparian." [13] Under Harmon's approach each State could alter the river as much as it wanted with neither being held accountable under international law for having passed the pollution threshold. Given the general directional flow of contaminating substances in moving waterways, the lower riparian is clearly placed in a disadvantageous position. At the other extreme, emphasis on the absolute territorial inviolability of the downstream State is to permit the lowest riparian to veto any use by the upper riparians, even if the stream is returned to something very close to its " natural " state before it reaches the border.[14] Further it allows the down-stream State to bring about any degree of change in the river which, since it

[12] Judson Harmon, United States Attorney-General, Dec. 12, 1895, in Moore, *International Law Digest*, Vol. 1, p. 654.

[13] A. P. Lester, " River Pollution in International Law " (1963) 57 A.J.I.L. p. 832.

[14] For good reason, such a position was found untenable in the *Lake Lanoux* Arbitration. See Brunson MacChesney, " *Lake Lanoux* Case " (1959) 53 A.J.I.L. 165–167.

then flows into the *res communis* of the sea, damages no territorial sovereign in the way necessary to serve as the basis for a claim.[15] From an ecological or economic perspective, neither extreme represents a rational approach to environmental pollution.

C. *Pollution as Damage*

Pollution as damage is a very wide category and to distinguish between the various approaches which have been taken within it one must keep in mind two interrelated questions: to what must the damage be done to be considered pollution and what level of damage is necessary? The most useful methodological distinction is made between damage to man and his property and damage to the environment.[16]

Damage to man and his property

The decision of the tribunal in the *Trail Smelter Arbitration* is a good illustration of a high threshold approach. Feeling limited by the *compromise* to a determination of the " extent to which *property* in the State of Washington has been damaged by fumes from Smelter at Trail, B.C.," [17] the tribunal, in effect, construed the term " damage " to include only " tangible injury translatable into provable monetary damages." [18] The tribunal did state that both international and United States municipal law deny any State " the right to use or permit the use of its territory in such a manner as to cause injury by fumes in or to the *territory* of another or the properties or persons thereof." [19] This implies the possibility of damage to the environment itself, at least as it exists as the property of the injured State. Yet the award of compensation under United States municipal law indicates that non-monetary damage to the evergreen population of Washington (as well as potential tourism and other aesthetic or

[15] See *Trail Smelter*, n. 17 *infra*.

[16] It must be pointed out that there are a number of other differences between various kinds of damage concepts which, because of the complexity of the subject and limitations of this paper, cannot be treated in any systematic way. The question of where the damage occurs, within or beyond the limits of national jurisdiction, is a very important factor, as is the part of the biosphere regulated by each concept of pollution. Further study might well be able to establish a correlation between all these factors and the actual definitions chosen.

[17] Trail Smelter Arbitral Tribunal, Decision, April 16, 1938, *UNRIAA*, Vol. 3, p. 1918. (emphasis added).

[18] Alfred P. Rubin, " Pollution By Analogy: The Trail Smelter Arbitration " (1971) 50 *Oregon Law Review* 272. The logic, or lack thereof, which the Tribunal employed in arriving at this decision is fully discussed in Professor Rubin's article. What is important, for our purposes, is the concept of what interests could or could not be considered as protected by that decision and thus the area within which the label " pollution " could be applied.

[19] Trail Smelter Arbitral Tribunal, Decision, March 11, 1941, *UNRIAA*, Vol. 3, p. 1965. (emphasis added).

speculative interests) did not meet the criteria of our threshold concept of pollution.[20]

Further, besides requiring that such damage actually occur before the threshold could be passed,[21] the tribunal said that the claim could not be accepted unless " the case is of serious consequence and the injury established by clear and convincing evidence." [22] It should also be noted that in the dispute no question arose of the legal effect of alterations to the biosphere outside the territorial jurisdiction of any State.

The 1909 Boundary Waters Treaty between the United States and Great Britain (Canada) also approaches pollution from a damage perspective. Article IV states, " It is further agreed that the waters herein defined as boundary waters and waters flowing across the boundary shall not be polluted on either side *to the injury of health or property on the other*." [23] Of particular significance, the determination of exactly what constitutes " injury " was not made clear in the Treaty but was left to the International Joint Commission (IJC) which it established. Although the word " polluted " implies that *any* alteration might serve as the basis of an international claim, the second part of the sentence makes it clear that the legal effects would attach only when such alteration involved " injury." The word of the IJC furnishes an important view of how this threshold has been defined. As will be shown in the next section, the damage context within which the IJC's findings are framed has resulted, at times, in a real discrepancy between the standards suggested by the Treaty and those actually employed by a body desirous of lowering the threshold of pollution.[24] In any case, the threshold envisioned by this agreement seems to lie markedly below that required under the *Trail Smelter* decision, despite an apparent stipulation that the injury be done to " health or property " on the other side of the boundary. What are missing are qualifying terms like " substantial," " serious," and " established by clear and convincing evidence."

An even lower threshold is implied by an exchange of notes in 1961

20 Rubin, " Pollution By Analogy," p. 273.

21 *Ibid.*

22 Trail Smelter Arbitral Tribunal, Decision, March 11, 1941, p. 1965.

23 Treaty Between His Majesty and the United States of America Relating to Boundary Waters and Questions Arising Along the Boundary Between Canada and the United States, Washington, January 11, 1909, *United States Treaty Series*, Vol. 548, Art. IV, p. 3. (emphasis added).

24 A similar approach was taken by France, Belgium, and Luxembourg in 1950. These countries set up a Tripartite Standing Committee on Polluted Waters and entrusted a technical subcommittee with the task of drawing up pollution standards. Unlike the IJC, however, the subcommittee was not confronted with a formal and established approach to the problem. Protocol Between France, Belgium, and Luxembourg to Establish a Tripartite Standing Committee on Polluted Waters, Brussels, April 8, 1950, *UNTS*, Vol. 66, p. 285.

between the United States and Mexico over the transboundary emission of fumes from American stockyards in El Paso to Ciudad Juarez. Although Mexico did make the claim that " serious physical and economic damages " were being suffered by the city's residents and businesses,[25] the United States' reply which, in apologetic tones, declared the problem solved, accepted responsibility for something far less serious than damage as used in the *Trail Smelter* sense. Regretting " any *inconvenience* or *discomfort* that the activities of the plants might have caused," the Secretary of State, Dean Rusk, explained the meteorological reasons for the problems and included with these odours " other pollutants such as smoke . . . and traffic dust " as contributing to the air pollution of the area.[26] Yet it was the odours against which Mexico had protested and the United States felt that the test of whether or not that problem had been taken care of was to see how the people who had originally complained felt after the changes had been made. Compensatable injury, " damages," were disregarded in the final stages of the correspondence as no reference was made to any decrease in the economic or physical damage of which Mexico had complained. It was said simply that people agreed that the " *objectionable* odours have been removed." [27]

Similarly, a report by Professor Hogger to the European Conference on Air Pollution stresses the need to include " inconveniences " as legally relevant forms of pollution damage.[28] Hogger also confronts directly the problem of defining the threshold of what degree of inconvenience to consider unacceptable, a problem which Charles Ross suggests is avoidable through a switch of emphasis from water quality to emission standards.[29] Yet, while emission standards may be a preferable policy tool for combating pollution, they still require some determination of the overall level of acceptable environmental change. This is not an easy task:

> It is, indeed, in the very nature of the problem that such limits can only be determined approximately. Apart from the difficulties of logic and methodology, there is a considerable linguistic problem, in the sense that there is no generally accepted definition of such terms as " inoffensive," " presumed," etc.[30]

Absolute purity is seen as an extremely costly and unnecessary basis

[25] Chargé d'affaires *ad interim* of Mexico (Gallardo) to the Secretary of State (Rusk), note, April 6, 1961, Whiteman, *Digest of International Law*, Vol. 6, p. 257.
[26] Secretary of State to the Chargé d'affaires *ad interim* of Mexico, note, Aug. 17, 1961, Whiteman, *Digest of International Law*, Vol. 6, p. 258.
[27] *Ibid.*, p. 259 (emphasis added).
[28] Professor Hogger, *Comparison of National Laws and Regulations, Possibility of Standardising These and Drawing Up Technical and Legislative Agreements and European Conventions on Air Pollution*, presented at the European Conference on Air Pollution, Strasbourg, June 24–July 1, 1964, pp. 439–440.
[29] Ross, *National Sovereignty*, p. 252.
[30] Hogger, *Comparison of National Laws*, p. 441.

for creating standards. Hogger implies that cost-benefit considerations should really predominate in the case of " inconveniences," while one might go to greater lengths and take net economic " loss " into consideration where actual damage was done or the health of populations endangered.[31] While this seems reasonable in situations where disruptive economic changes might bring only minor improvement of a problem of limited proportions, the line between " inconvenience," " damage," and " danger " becomes an important one to draw and Hogger gives little indication of how this should be done.

Although one may question their value as useful sources of international law, several judicial decisions made within the United States municipal legal system do help illustrate the range of concepts from high to low thresholds of damage to man and his property.[32] In *Missouri* v. *Illinois* (1906), Justice Holmes emphasised the need for a case to be " of serious magnitude, clearly and fully proved," [33] before the court should intervene, a concept of great importance to the *Trail Smelter* tribunal. The inability of Missouri to draw a clear connection between its increased rate of deaths due to typhoid and the Chicago sewage resulted in the case being dismissed.[34] New York also lost its case against New Jersey in 1921 when it could not show that New Jersey's proposed sewage disposal plan " would so corrupt the waters of the Bay as to create a public nuisance by causing offensive odours or unsightly deposits on the surface or that it would seriously add to the pollution of it." [35] Despite some confusion with the separation of public nuisance from an undefined " pollution " concept, the court did imply that " offensive odours " and " unsightly deposits " could be made the basis of a valid claim if sufficiently proved.[36]

Sierra Club v. *Morton* (1972), while dismissed because the Sierra

[31] *Ibid.*, p. 442.

[32] No attempt is made to accept or refute the view of some publicists that much of the unwillingness of the Supreme Court justices, particularly Holmes, to impose a low threshold view was due less to limited definitions of pollution than to a very real concern for the role of the Supreme Court in resolving interstate disputes. See Rubin, *Pollution By Analogy*, pp. 269–271. For whatever reasons they decided as they did, the conceptions of pollution which emerge from those decisions are useful illustrations of the differences previously examined.

[33] *Missouri* v. *Illinois and the Sanitary District of Chicago*, 200 U.S. 521 (1906).

[34] *Ibid.*, pp. 525–526.

[35] *New York* v. *New Jersey and Passaic Valley Sewerage Commissioners*, 256 U.S. 313 (1921).

[36] *Ibid.* The court, in effect, set up two thresholds of unacceptable environmental change. One which it called " pollution " was measured in terms of the change in the percentage of dissolved oxygen in the water. The other involved a question of public nuisance with an unarticulated degree of " unsightliness " or " objectionability " serving as the extremely vague basis for determining what was unacceptable. New Jersey's proposed plan was viewed as bringing about effects which transcended neither threshold and implicit was the belief that, by remaining within the legal limits established by these thresholds, " injury " would not be done to the Bay.

Club was unable to demonstrate the injury which would be done to its members by a proposed Disney development in a secluded mountain site, left open the possibility of an individual being able to claim injury to himself on the basis of aesthetic and ecological changes made in the environment which might not be easily translatable into economic terms. The court stated that " aesthetic and environmental well-being, like economic well-being, are important ingredients of the quality of life in our society " and quoted the *Data Processing* case (399 U.S. 154) in arguing that " the interest alleged to have been injured ' may reflect " aesthetic, conservational, and recreational " as well as economic values.' " The court objected not to the idea that such interests could be injured in a legal sense but to the fact that the Sierra Club had not shown the connection of the injury to the individuals it represented.[37]

Damage to the environment

Reacting to the limitations placed on a concept of pollution restricted to damage to specifically " human " interests, some publicists and organisations have moved to considering damage to the environment itself as legally significant. Justice Holmes, in *Georgia* v. *Tennessee Copper Co.* (1906), saw the claim as based on injury done to Georgia as a quasi-sovereign. Yet the damage was done to the environment which was simply being defended by the State that had territorial jurisdiction over it.

> It is a fair and reasonable demand on the part of a sovereign that the air over its territory should not be polluted *on a great scale* by sulphurous acid gas, that the forests on its mountains... should not be further destroyed or threatened by the act of persons beyond its control.[38]

The evidence of severe damage was so substantial " as to make out a case within the requirements of *Missouri* v. *Illinois*," [39] and Georgia won the case.

The Brussels Convention Relating to Intervention on the High Seas in Cases of Oil Pollution Casualties, although clearly reflecting the active measures resulting from its potential application, also sets a relatively high threshold of damage to the environment. Parties are allowed to act only against " *grave and imminent danger* to their coastline or related interests from pollution of the sea by oil, *following* upon a maritime casualty or acts related to such a casualty, which may reasonably be expected to result in *major harmful* conse-

[37] *Sierra Club* v. *Morton*, 405 U.S. 734–735 (1972).

[38] *Georgia* v. *Tennessee Copper Co. and Ducktown Sulphur, Copper and Iron Co.*, 206 U.S. 238 (1906) (emphasis added).

[39] *Ibid.*, p. 239.

quences." [40] On the other hand, a joint communiqué released by the United States and Canada, which significantly extends to "areas beyond the limits of national jurisdiction" as well as the territory of other States, speaks of the international responsibility of States not to do damage to the environment. The term "damage" is unqualified and suggests a lower threshold than the above convention.[41]

A number of terms have been chosen to connote varying degrees of environmental damage. The presence of "harmful" or "potentially harmful" substances in the environment has been suggested by Dr. Herbert Volchok as a useful basis for pollution but, as he admits, it is really the "relative levels" of harm to the environment that are important from a legal standpoint. Dr. Volchok does not elaborate on what these levels might be although he suggests that they should be much lower than is generally believed.[42] The term "harmful" is also employed in the 1967 Outer Space Treaty. Article 9 requires parties to carry out their exploration of celestial objects in such a way as "to avoid their *harmful contamination* and also *adverse changes* in the environment of the Earth resulting from the introduction of celestial matter." [43] Neither of the italicised terms is further defined within the Treaty but relatively precise guidelines were set up in 1964 by the Consultative Group of the Committee on Space Research (COSPAR) of the International Council of Scientific Unions to prevent the undesired contamination of outer space and were basically adhered to in the United States explorations near Mars. Efforts have also been made to limit, as much as possible, effects on the Earth's environment from returning missions, yet here as well clear-cut standards of actual acceptable damage to the environment have not been worked out.[44]

The United States National Oceanic and Atmospheric Administration (NOAA) report to the Congress on ocean dumping defines ocean pollution as "the *unfavourable alteration* of the marine environment... through direct or indirect effects of changes in energy patterns, radiation levels, chemical and physical constitution and dis-

40 International Convention Relating to Intervention on the High Seas in Cases of Oil Pollution Casualties, Brussels, Nov. 29, 1969, *International Legal Materials*, Vol. 9, No. 1 (1970), Art. I (1), p. 25 (emphasis added).

41 United States-Canadian Joint Communiqué, Train and Davis, issued on July 14, 1972, in *The International Law of Pollution*, James Barros and Douglas M. Johnston, comp. (New York: Free Press, 1974), p. 345.

42 Herbert Volchok, statement before U.S. Senate Committee on Commerce, Sub-committee on Oceans and Atmosphere, 93rd Congress, 1st session, July 12, 13 and 28, 1973, *Ocean Pollution*, pp. 52–59.

43 Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies, Washington, London, and Moscow, Jan. 27, 1967, *UNTS*, Vol. 610, Art. IX, p. 210 (emphasis added).

44 Frederik L. Kirgis, "Technological Challenge to the Shared Environment: United States Practice" (1972) 66 *A.J.I.L.* 309–310.

tribution, abundance, and quality of organisms." [45] Apparently including such problems as overfishing and spoliation in general, this definition also makes explicit the fact that pollution need affect man only in the most indirect way, perhaps simply by lessening his enjoyment of nature. When faced with the problem of what to consider " unfavourable " for purposes of regulation, the NOAA report says most standards are not strict enough. In particular, it argues that many of these standards ignore negative long-term effects of sub-lethal doses of substances like DDT on the entire ecology of a given ocean area, looking instead for the concentrated, acute effects on particular organisms.[46]

An even more inclusive view of pollution is advanced by Alfred Rubin who, within a damage context, would define damage to mean " any alteration in the predicted availability of natural resources." [47] Offered in a discussion of the *Trail Smelter* decision, Professor Rubin's definition was intended to allow for changes made directly on the environment for which one would not be able to assign monetary " damages." While it takes into account the sort of long-range effects that worried the NOAA, this view would also seem to label " pollution " the exploitation of any non-renewable resource. As important as this might be to frame one's thinking about the relationship between man and his environment, or, more practically, to set up a procedure requiring proof of " acceptable damage " before any activity which would have an effect on the environment could be started, it would be difficult to apply such a concept as a final legal standard.

Many of the problems in the damage approach are found where, due to lack of agreement, limited perspective, or the strict legal consequences envisioned for attaching the label, a high threshold has been chosen. Requiring " serious " or " grave " damage to make an alteration legally significant may result in the overlooking of gradual changes with perhaps even greater long-term impact. Where restricted to " damage " as tangible injury of a monetarily determinable nature, the damage viewpoint may refuse protection to much of the " unowned " area and resources of the world. At the other extreme, very low levels of acceptable damage could, depending on the legal consequences, demand a costly and relatively unrewarded " cleaning-up " operation. Emphasis on damage, unless very widely defined, offers little positive guidance about man's interaction with the environ-

[45] U.S. Department of Commerce, National Oceanic and Atmospheric Administration, " Report to the Congress on Ocean Dumping and Other Man-Induced Changes to Ocean Ecosystems (March 1974), p. 6 (emphasis added).

[46] *Ibid.*, p. 20. The report says most tests look for the dose required to kill 50 per cent. of the test organisms.

[47] Rubin, *Pollution By Analogy*, p. 292.

ment. It stresses a series of levels and degrees of prohibition, rather than providing international law with a progressive and co-operative approach to the problem of maintaining the integrity of natural systems while allowing for social and economic development.

D. *Pollution as Interference with Other Uses of Environment*

Another approach which has received substantial support focuses on pollution as that which interferes with the various uses to which a particular environment is or can be put. Largely the product of international agreements regulating busy waterways and ocean areas, this category has been characterised in most of its forms by a very short-sighted, anti-ecological bias. The environment is important only to the degree that it is useful to man's immediate interests and environmental alteration is something to be halted only if the benefits of so doing, increased efficiency or expansion of usage, outweigh the costs. The threshold concern here is the degree of interference necessary for pollution to have occurred, while a second major distinguishing feature between these views is the various uses with which it is legally significant to interfere.

The Frontier Treaty of 1960 between the Netherlands and the Federal Republic of Germany adopts a use orientation. Article 58 (2) (e) requires both parties to take steps " to prevent such *excessive pollution* of the boundary waters as may *substantially impair* the *customary use* of the waters by the neighbouring State." [48] The " customary use " orientation of this agreement is retained, although the interference threshold lowered somewhat, by the Indus Waters Treaty of the same year. Pakistan and India thereby agreed to refrain from " undue pollution . . . which might adversely affect uses similar in nature to those to which the waters were put on the effective date." [49]

The rationale behind the use approach is spelled out well by Justice Holmes in the 1931 case of *New Jersey* v. *New York*:

> A river is more than an amenity, it is a treasure. It offers a necessity of life that must be rationed among those who have power over it... Both states have real and substantial interests in the river that must be reconciled as best they may be. The different traditions and practices in different parts of the country may lead to varying results, but the effort is always to secure equitable apportionment without quibbling over formulas.[50]

[48] Frontier Treaty Between the Kingdom of the Netherlands and the Federal Republic of Germany, the Hague, April 8, 1960, in force Aug. 1, 1963, in *The International Law of Pollution*, p. 158 (emphasis added).

[49] The Indus Waters Treaty (1960), Karachi (Sept. 19, 1960), *The Indian Yearbook of International Affairs* (1960–1961), Vol. IX–X, p. 364.

[50] *New Jersey* v. *New York* 283 U.S. 342–343 (1931).

The court used the services of a technical expert to see what the results of New York's proposed water diversion plan would be in terms of New Jersey's uses of the Deleware watershed and found that the planned 600 million gallons per day diversion would produce somewhat serious effects on recreation and oyster fishing uses. The decision was thus to limit New York's withdrawal to 440 million gallons daily.[51] Here a *new* use of the river's waters was integrated into an overall picture of interdependent, though potentially conflicting interests.

A study prepared in 1958 for the Economic Commission for Europe recognised that every State on an international river has the right to use the river as a natural resource. Although no lower riparian could expect to receive the water in a completely natural condition, " it is accepted that each upstream country has a duty to pass on the water in a usable condition." [52] " Usable condition " was described in the following paragraph as being equal to the condition in which the water was maintained in practice in the upstream State. The minimum acceptable condition, however (described as " wholesome ") would be that the water contain " significant amounts of dissolved oxygen for at least 95 per cent. of the year." [53]

An interesting contrast to this view of the use approach is provided by a report to the same organisation made one year earlier by the Group of Experts consulted by the Executive Secretary of the ECE. Where the 1958 study spoke in relatively specific terms of what minimal standards are for a " usable " river, this report gave no such criteria and even stated that " although the discharge of, for instance, salt, to any water is theoretically pollution, if it took place to a river whose sole use was navigation then it would not qualify to be classified as pollution." [54] Neither provided an effective guarantee for future uses of the river, but the 1958 study at least showed some concern that the river be maintained above a minimal level of purity. On the other hand, where the 1957 report called pollution any change which adversely affected the existing uses of the downstream State,[55] the 1958 study required that the water be passed on in usable condition. The point may be trivial, but there seems to be a difference between the two views as to whose " use criteria " would determine whether the river was polluted.

A third definition offered to the ECE declared a river to be polluted when its water is " less suitable for any or all of the purposes to

[51] *Ibid.*, p. 50.
[52] Economic Commission for Europe, " A Study on Water Pollution Control Problems in Europe," Feb. 26, 1958, Whiteman, *Digest of International Law*, Vol. 3, p. 1049.
[53] *Ibid.*
[54] *Ibid.*, p. 1043.
[55] *Ibid.*

which it would be suitable in its natural state." [56] While this definition offers no more precise a view of the threshold at which changes would be legally significant, it clearly requires that a waterway be preserved at the highest level of purity necessary for any potential use, even if not practised by anyone at the time.

Among publicists, Charles Bourne is a firm supporter of the use approach. " Man's concern is not whether the available water is pure, but whether there is enough of usable quality to satisfy his needs." [57] For Professor Bourne, sewage disposal is just one more possible use for a waterway, a use which will have a higher or lower priority depending on the needs of those who control it. Levels of cleanliness thus have significance only in their relationship to the requirements of other uses considered valuable by the community.[58]

Professor Bourne speaks highly of the 1966 Helsinki Rules of the International Law Association (ILA) which do articulate this viewpoint in a persuasive context. After a period of soul-searching in Article IX, where the ILA stresses the need to adopt a definition of pollution which recognises injury to the environment " irrespective of its effects on subsequent users," [59] Article X makes it clear that " equitable utilisation " is the key concept when it comes to determining the actual legal threshold of pollution. Pollution which results in injury should be seen within the " overall perspective of what constitutes an equitable utilisation " [60] and thus what the ILA considers pollution for definitional purposes may not be illegal if the use which causes it is in keeping with the equitable utilisation principle.[61] The Helsinki Rules also limit the ability of a State to argue in favour of preserving a river for future use. Article VII says: " A basin State may not be denied the present reasonable use of the waters of an international drainage basin to reserve for a co-basin State a future use of such waters." [62]

A final example which must be subsumed under the use category, although it reflects a far greater concern for the natural world, is the Canadian " Arctic Waters Pollution Prevention Act." The word " pollution " is largely avoided but the Act's definition of " waste " is aimed at the threshold with which this paper is concerned. Canada's

56 *Ibid.*

57 Bourne, C. B., " International Law and Pollution of International Rivers and Lakes " (1971) *University of British Columbia Law Review*, p. 117.

58 *Ibid.*, p. 118.

59 International Law Association, Helsinki Rules on the Uses of the Waters of International Rivers, Helsinki, August, 1966, in *Report of the Fifty-Fifth Conference* (London, 1967), p. 496.

60 *Ibid.*, p. 499.

61 *Ibid.*, p. 500. This is somewhat qualified by a reference to " substantial " injury and it remains unclear whether this substantial injury, even if caused by a valid use of the water would be a legal delict by the injuring State.

62 *Ibid.*, p. 492.

jurisdiction is extended over a 100-mile zone to regulate the emission of any substance or water that "would, if added to any waters, degrade or alter or form part of a process of degradation or alteration of the quality of those waters to an extent that is detrimental to their use by man or by any animal, fish, or plant that is useful to man." [63] While this definition permits Canadian authorities to react directly against pollution which may affect a given species of fish far more directly than it does Canada's use of the seas, there still remains some doubt as to the fate of living organisms that may not be considered "useful to man."

As was stated earlier, the major objection to this approach is the disregard for the biological characteristics of ecological systems that it almost always represents. It is so directed to the balancing of State and individual interests that many important components of the biosphere may never be considered. This is particularly true if the question becomes reduced to a simplistic cost-benefit analysis into which it may be extremely difficult to fit the value of an unexploited species of fish There is an economic rationality to the approach, similar to the "polluter-pays" principle put forth by the Organization for Economic Co-operation and Development,[64] which seems to ignore the very real limits of the planet's ability to absorb man-caused waste and other environmental changes.

In adopting a use orientation, the ILA and others have attempted to incorporate the "reasonable man" test into environmental planning and dispute settlement procedures. Yet "reasonable" people do now and will continue to disagree over the priority of uses for a river, for example. However successful the reasonableness standard may be in a vertically-structured municipal law context where some form of judicial authority is able to weigh these opposing views and render a binding decision, the horizontal nature of the present international legal system makes the use of this standard far less effective on an international level.

Further, while one presumes that the ILA would be hesitant to endorse the extreme "full utilisation" approach taken by Myres McDougal and Norbert Schlei in their defence of U.S. nuclear testing in the Pacific Ocean, "The Hydrogen Bomb Tests in Perspective: Lawful Measures for Security" does reveal the danger inherent in relying solely on reasonableness in formulating legal norms to protect the international environment. Viewing the "technical prescriptions" of such concepts as the "territorial sea," "freedom of the seas," and

[63] "Arctic Waters Pollution Prevention Act," Canada (1970) 9 I.L.M. 544.

[64] Organisation for Economic Co-operation and Development, "Guiding Principles on the Environment," May 26, 1972 (1972) 11 I.L.M. 1172.

so forth as " highly flexible policy preferences " to be manipulated by decision makers in a particular controversy, McDougal and Schlei assert that, " for all types of controversies the one test that is invariably applied by decision makers is that . . . standard of what, considering all relevant policies and all variables in context is *reasonable* as between the parties." Naturally, " security " interests have a high priority for most decision makers and the hydrogen bomb tests must be seen as " reasonable, and hence lawful, within the régime of the high seas." [65] If such a " use " of the environment can be considered " reasonable," at least by two respected publicists, what cannot? All too often " in light of all relevant factors," and similar phrases used to justify a " reasonable use " approach, in fact have resulted in a position of secondary importance for environmental " variables."

E. *Pollution as Exceeding Assimilative Capacity of Environment*

The type of concern just mentioned, the growing realisation of the frailty and interdependence of the components of the biosphere, has brought about another view of pollution as waste which, either in quantity or in kind, is such that it cannot be broken down and rendered harmless by natural processes. Barry Commoner spoke to the Senate Commerce Committee's Subcommittee on Oceans and Atmosphere about pollution as the " overloading of natural cycles." " Human beings pollute the enivronment because their activities have broken out of the closed cyclical network in which all other living things are held." [66] One of the most important of these " activities " has been the development of synthetic chemicals such as PCBs which, as C. I. Jackson points out,[67] were created and their use made widespread long before their effects on the environment were known. " Foreign " to the natural world and incapable of being broken down by normal ecological processes, their very existence may be " polluting " the planet.

Determining the existence of legally significant pollution primarily on the basis of assimilative criteria has received some support from publicists, particularly those concerned with the need to encourage developing countries to take a direct interest in the pollution problem. Rubin writes of the value of a clear understanding " that pollut-

[65] Myres S. McDougal and Norbert A. Schlei, " The Hydrogen Bomb Tests in Perspective: Lawful Measures for Security," in Myres S. McDougal and Associates, *Studies in World Public Order* (New Haven: Yale University Press, 1960), pp. 775–778.

[66] Barry Commoner, statement before U.S., Senate, Committee on Commerce, Subcommittee on Oceans and Atmosphere, 92nd Congress, 2nd session, Oct. 18 and Nov. 8, 1971, *International Conference on Ocean Pollution*, p. 77.

[67] C. I. Jackson, " Dimensions of International Pollution " (1971) 50 *Oregon Law Review* 242.

ing a neighbour's territory, befouling the seas, or despoiling the environment of one's own territory so as to endanger the ecology of the planet constituted a violation of an international obligation." [68] His view of the assimilation concept clearly transcends political boundaries giving international law purview over changes within a State's territory which may not result in provable "damage" to interests outside but do "endanger" the Earth's ecology. Ross refers to Canada's concern within the IJC that the United States' recent move from water quality to emission standards heralds an attempt to limit industrial expansion along the Great Lakes by setting extremely high international effluent standards which do not reflect the actual assimilative capacity of the lakes. Once developed, the United States appears unwilling to permit other countries to use the same development methods it did and the costs of "unnecessary" United States-sponsored pollution control seem prohibitive. "The very essence of assimilative capacity suggests certain rights or equitable entitlements," [69] and developing countries need to have some sort of "right" to alter the environment for development purposes. Nevertheless, while recognising the need to take the particular characteristics of each body of water into account in setting up water quality objectives for it, the IJC said in its report on Rainy River that such objectives "should not . . . tolerate the maximum quantity of domestic and industrial wastes that the stream can assimilate." Exactly what role assimilative criteria should play was left unclear.[70]

However useful this view may be to those concerned with either the very real problems faced by the environment on a global level or the requirements of developing countries, this approach is insufficient in itself as the basis of an international law concept of pollution. First, there is the problem of data. It is difficult to determine with any certainty what constitute the assimilative limits of a given environment. This is particularly true in the case of substances like PCBs whose environmental effects cannot be known immediately. Similarly, what may appear to be acceptable limits of a substance in terms of its detrimental effect on the ecological structure of a swiftly-flowing river may result in sizeable deposits in the ocean into which it empties and seriously disrupt the maritime ecology. To translate marginal changes in as vast a system as the world's environment back into useful standards of pollution on a local level is a difficult task.

On another level, there is a great deal of uncertainty as to when

68 Rubin, *Pollution By Analogy*, p. 275.
69 Ross, *National Sovereignty*, p. 254.
70 International Joint Commission, "Report on the Pollution of Rainy River and the Lake of the Woods" (1965), in *The International Law of Pollution*, p. 104. See *infra*, p. 553 for further discussion of this report.

it can be said that actual damage has been done to the integrity of natural cycles. Is the process of eutropiphication taking place in some of the United States' most "polluted" lakes a "natural" process, simply responding to the introduction of human wastes, or is it a sign that the ecological systems of the area have been violated? If the latter, at what point should the law intervene? Before eutropiphication begins, at some level of "unacceptable" eutropiphication, or when the eutropiphication itself affects a wider circle and cannot be absorbed into the ecology?

Finally, one must object to the assimilative approach in that the threshold it seems to advocate offers little or no protection to aesthetic values and other interests which could be irreparably damaged by environmental changes that have no significant effect on particular ecological systems. The towering office building that dwarfs and disgraces the structures around it and the hamburger stands and souvenir shops which increasingly obscure man's view of and contact with nature may have little or no effect on the functioning of any natural cycle and yet constitute environmental alterations many would find unacceptable. At the same time, a sufficiently narrow assimilative approach might label the existence of a local garbage dump "pollution" since the wastes deposited there either could not be integrated "naturally" into the land or had a serious impact on the types of animal and plant life which the dump area could support.

CONCLUSION

This paper has presented five ways of approaching the question of what is legally significant pollution. Of the last three, none is completely independent of the others. Within the use concept there are assumed views of what constitutes "damage" just as the damage approach recognises, to some degree, that damage to another use of the environment may be pollution. Damage to particular aspects of the environment and exceeding the assimilative capacity of a given environment are clearly related. The ability of the learned Justice Holmes to represent both the use and damage approaches in different contexts is further evidence of their inter-connection. In some cases, it is a matter of emphasis. Yet no consistent view of pollution emerges from this past record. The differences between definitions are more than semantic; they reflect inconsistent views of what kinds and degrees of environmental change men should be allowed to make.

That is the key issue. Having decided upon a "pollution" threshold, different bodies have then chosen a variety of means to see that this threshold was made legally enforceable. Some prohibited any discharges of certain kinds or attempted to regulate discharges to keep their effects within acceptable limits. Others decided to permit discharges until they reached the chosen threshold and then to attach legal consequences designed either to encourage potential polluters to refrain from undesired environmental degradation or to force them, at least, to pay some kind of compensation for their harmful actions. Yet this becomes a matter of tactics: how, in a specific case, can man and his environment best be protected from the adverse effects of societal development? First, one must find a satisfactory way to determine the " pollution " threshold.

II. CONTEMPORARY DEFINITIONAL TRENDS

Given the confusion that the relatively short history of State practice and uncertain " general principles " of international law have created with regard to pollution, recent efforts of international bodies to implement a meaningful and useful pollution concept are instructive of the problems faced in arriving at a general definition for the future.

A. *Label Substitution*

The lack of precision which has become characteristic of the term " pollution " has led in many cases to the invention of new labels. The Canadian " Arctic Waters Pollution Prevention Act " and several of the IMCO conventions declare in their titles the intention of preventing pollution. Yet none defines pollution within the text. As an example, the International Convention for the Prevention of Pollution from Ships concerns itself with the discharge of " harmful substances " from ships, with " harmful substances " defined as " any substance which, if introduced into the sea, is liable to create hazards to human health, to harm living resources and marine life, to damage amenities or to interfere with other legitimate use of the sea." [71] This concept varies little from that put forth by other recent international agreements which employ an expanded view of pollution. Similarly, the Canadian legislation, by employing the term " waste " within a use approach, is still confronted with the problem of deciding when a given level of interference becomes illegal. The fact that Canadian authorities can set up relatively precise levels for determining a " waste " threshold is a function not of the added precision of the new term, but of the relative ease in reaching agreement on standards within a single national government.

Although this label substitution makes it no easier to describe with precision the *level* of environmental degradation considered illegal,

[71] International Conference on Marine Pollution: International Convention for the Prevention of Pollution from Ships, London, Nov. 2, 1973 (1973) 12 I.L.M. p. 1320.

it can be useful in narrowing the *scope* of environmental effects to be regulated. " Waste " can be more easily perceived as simply a form or subset of a larger concept called " pollution." This can also work the other way, as the Scandinavian Convention on the Protection of the Environment illustrates. There the primary concern is for " environmentally harmful activities " which include air and water pollution as particular types but set pollution, as such, apart from other problems like " sand drift," " vibration," and " ionising radiation." [72] This distinction is a useful one for clarifying what is meant by pollution.

B. *Combination of Approaches*

Another development, which is directed more at the threshold problem of what polluting activity is legally significant has been an increasing tendency for international groups to adopt definitions of pollution which embody two or more of the approaches described in the first section. At the 58th Conference of the International Law Association, " Draft Articles on Marine Pollution of Continental Origin " were presented which, coming from the body responsible for the relatively " use-oriented " Helsinki Rules, provides a good example of this development. After adopting a descriptive definition of pollution in Article I, the draft turns to the actual legal obligations of a State. Article II de-emphasises the equitable utilisation principle of the Helsinki Rules and, while making a distinction between the legal consequences of existing and future pollution, says that a State " shall prevent any form of existing sea-water pollution . . . which would cause substantial injury in the territory of another State or to any of its rights under international law or to the marine environment." [73] Inter-State injury receives a more prominent place and an obligation to the environment itself is put forth for the first time. " There is now more and more support for the philosophy according to which maritime waters need general protection, as they constitute . . . a sort of global resource, and the flora and fauna living in these massive quantities of waters must be protected in the interest of all." [74] Article III stresses the need for international standards for seawater pollution to be established and, with the help of a group of technical experts, sets up a list of " relevant factors " to be considered. This list takes into account a wide variety of monetary, recreational, conservation, and ecological interests, clearly not limited by a use con-

[72] Denmark-Finland-Norway-Sweden: Convention on the Protection of the Environment, Stockholm, Feb. 19, 1974 (1974) 13 I.L.M. 591.

[73] International Law Association, " Draft Articles on Marine Pollution of Continental Origin," *Report of the 58th Conference* (New York, 1972), Art. II, p. 100.

[74] *Ibid.*, Comment on Art. II, p. 101.

cept, and carefully refrains from assigning any relative importance to a particular factor over others.[75]

The IJC has gone through a similar period of transition as it has confronted a series of pollution questions along the United States-Canadian border. Oriented by the 1909 Treaty and subsequent U.S.-Canadian references to the IJC to view pollution as " injury of health or property " on either side of the border, the Commission has, at times, felt it appropriate to fit into this damage framework some aspects of what it considered pollution that seemed to stretch somewhat the pollution concept of the Treaty. The report on Rainy River contains two good examples. First, the actual existence of harmful conditions in the river is established solely by the use of various scientific standards, such as coliform concentration and dissolved oxygen content. While this evidence of " pollution " of the water is not said to be legally sufficient, the case appears to have been decided by applying these criteria. The evidence of actual injury to property is given little emphasis and no monetary value and, otherwise, much of the IJC's concern is for the river's unfit condition for bathing, recreation, and some forms of aquatic life.[76]

A second example is provided by the water quality objectives set up for the river. The importance of competing uses is clear as the " general objective " states that any waste discharged into the river should not " create conditions which will adversely affect the uses of these waters as a source of domestic or industrial water supply or for navigation, fish and wild life, bathing, recreation, agriculture, and other riparian activities." [77] The Commission seeks " to achieve the highest quality consistent with maximum benefit to all users." [78] Yet in the same section, which so clearly endorses a form of the use concept as a valuable index of pollution, the IJC still seems to find it advisable to view injury to the fish population of boundary waters within the terms of the Treaty. " Undue interference with the development of *desirable* types of aquatic life constitutes an *injury to property* even though it may be the property of the *public at large*." [79] One with an ecological bent might ask what type of aquatic life could be considered " undesirable," but the concept of freely swimming fish in an international waterway, fish that men may not like to eat or even catch, being claimed as property identifiable to one country or the other is particularly hard to rationalise and must be granted the status of " legal fiction " to mesh with the terms of the 1909 Treaty definition.

[75] *Ibid.*, Art. III (c), p. 102.
[76] IJC, *Report on Rainy River*, pp. 95–101.
[77] *Ibid.*, p. 105.
[78] *Ibid.*, p. 104.
[79] *Ibid.* (emphasis added).

Two important international dumping conventions, containing almost identical definitions of pollution, further illustrate the shift towards combining approaches. Article 1 of the 1972 Oslo Convention for the Prevention of Marine Pollution by Dumping from Ships and Aircraft requires all parties to prevent pollution by substances that " are liable to create hazards to human health, to harm living resources and marine life, to damage amenities or to interfere with other legitimate uses of the sea." [80] In addition to showing concern about damage to both humans and the environment and interference with other uses of the ocean, the Convention recognises " that the ecological equilibrium . . . of the sea [is] increasingly threatened by pollution." [81] The threshold problem is avoided somewhat by prohibiting any discharge of the substances considered most dangerous and requiring permits for the dumping of others. The only real difficulty here is in drawing up the lists. Yet, as is true of other " purity " oriented conventions, the Oslo Convention relies for its success on the relatively limited and controllable sources of the pollution to be prevented.

The 1972 Helsinki Convention on the Protection of the Marine Environment of the Baltic Sea, covering a larger range of problems, adopts a definition similar to that accepted at Oslo, expanding it somewhat to include the effects of the introduction of energy as well as substances. Eutropiphication also receives treatment as a type of " land-based pollution " and there is an explicit call for the development of uniform pollution control criteria. Once again, a list is drawn up of specific substances not to be introduced into the Baltic Sea area. Given the vague threshold and all-inclusive scope of pollution, as defined in Article 2, what is really meant by pollution is better discerned from the Annexes than the body of the Convention.[82]

Embracing similar concepts and yet lacking such explicit Annexes, the 1972 Stockholm Declaration on the Human Environment serves as a useful document to examine more carefully the implications of a widely-defined view of pollution. Principle 6 of the Declaration clearly accepts a role for assimilative criteria:

> The discharge of toxic substances or of other substances and the release of heat, in such quantities or concentrations as to exceed the capacity of

[80] Convention for the Prevention of Marine Pollution by Dumping from Ships and Aircraft, Oslo, Feb. 15, 1972 (1972) 11 I.L.M. 262. See also Convention on the Prevention of Marine Pollution by Dumping of Wastes and Other Matter, London, Nov. 13, 1972 (1972) 11 I.L.M. 1293.

[81] *Ibid.* " Annex I " listing the most dangerous substances specifically takes note of the ocean's assimilative capacity by excluding those types of a particular compound " which are rapidly converted in the sea into substances which are biologically harmless ": p. 265.

[82] Convention on the Protection of the Marine Environment of the Baltic Sea Area, Helsinki, March 22, 1974 (1974) 13 I.L.M. 544–557.

> the environment to render them harmless, must be halted in order to ensure that serious or irreversible damage is not inflicted upon ecosystems.[83]

Principle 7 goes on, with respect to the oceans, to add damage and use concepts in a form easily recognisable in conventions adopted since the Stockholm Conference. " States shall take all possible steps to prevent pollution of the seas by substances that are liable to create hazards to human health, to harm living resources and marine life, to damage amenities or to interfere with other legitimate uses of the sea." [84] Principle 21 makes explicit the existence of a relationship between the right of each State to exploit its own natural resources and " the responsibility to ensure that activities within their jurisdiction or control do not cause damage to the environment of other States or of areas beyond the limits of national jurisdiction." [85]

The combination of approaches outlined in the first chapter is evident. Yet the way they are separated and employed in the Stockholm Declaration reveals a view of pollution that does not simply " throw together " past definitions but tries, perhaps unconsciously, to select the most appropriate approach for the various levels and types of adverse environmental changes faced by the world, within the very real limiting context of what would be acceptable to the national delegations. The assimilative perspective is employed in Principle 6 in terms of an overall commitment to the integrity of the biosphere, a commitment which transcends and penetrates national boundaries. Not intended as the basis for setting up specific pollution standards, it is, at least, a clear reminder of the need to set an upper limit to " discharges " which threaten fundamental changes in the ecological balance.

The emphasis on ocean pollution in Principle 7 reflects, as Louis Sohn points out, the strong concern of many nations for the deteriorating condition of that particular part of the biosphere.[86] The vague concept of damage to " amenities," seen in other conventions, is included and there is the clear intention to allow almost any kind of detrimental environmental change in the oceans to fall within " pollution," given a degree of seriousness that is only generally hinted at by terms like " damage," " harm," and " hazards." Principle 21, far from implying a move toward near absolute sovereignty of a State over the management of its environment (as Sohn suggests),[87] sets out one aspect of the dual nature of the State's relationship with the

83 *Stockholm Declaration on the Human Environment*, p. 4.
84 *Ibid.*, Principle 7, p. 4.
85 *Ibid.*, Principle 21, p. 7.
86 Louis Sohn, " The Stockholm Declaration on the Human Environment " (1973) 14 *Harvard International Law Journal*, 463–464.
87 *Ibid.*, p. 492.

environment within its territory. The right to manage its own resources requires a corresponding responsibility that those resources be managed in such a way as to prevent damage to the environment outside of its territory. This is not just a throwback to earlier conceptions of pollution as a question of the territorial integrity of a sovereign State, although it does include that part of the more "traditional" view of legally-significant pollution. Within the framework of the other principles, it simply states one more area of international purview over the actions of States with regard to the environment.

C. *For the Future*

If any consensus emerges from what has been described, it is one which is evolving towards the use of a wide variety of *types* of criteria for defining pollution. The use, damage and assimilation approaches are being employed together to expand the range of effects from man-induced environmental change which can be judged, on the basis of some threshold of damage or interference, to fall under a legally significant concept of "pollution." What kinds of changes may be considered "pollution" is a question of scope which is approached only rarely and indirectly, as in the Scandinavian Convention which, as noted above, sets pollution apart from sand drift and vibration.[88]

The viability of a general concept of pollution for international law eventually depends upon the degree to which one can arrive at international agreement on the organisation(s) and procedure(s) which will identify both human activities and their by-products which cause unacceptable environmental change, and the levels of environmental change considered unacceptable. Much recent practice and the reasoned opinions of many publicists point to the need for precise threshold values of what this unacceptable level is, to be determined case-by-case given basic working limits, by groups of highly trained experts in the field. As has been shown, organisations and commissions have either performed or required the services of these groups to decide what is and what is not legally significant pollution, what demands regulation and what does not.

A general concept of pollution must include a means for determining these thresholds and the Stockholm Conference took a step in this direction by recognising the related problems of identifying and setting

[88] This question is one which has received scant and indirect treatment within this paper as well. For reasons of space and coherence, as well as lack of source material, I have limited myself to the range of effects and the threshold of damage or interference as the key variables distinguishing past definitions. The "scope" approach just mentioned seems to offer little useful in isolating the basic trends of current thought about what pollution is. Whether pollution includes overfishing, for example, is a difficult question to answer from past practice and one not usually considered central to the views on what pollution is.

limits for pollutants " of broad international significance " and recommending procedures for these purposes. In the section entitled " Pollution Generally," the Conference, while relying heavily on the co-operation of governments with international organisations, called, in Recommendation 74, for the development of a capability within the United Nations system " to provide awareness and advance warning of deleterious effects to human health and well-being from man-made pollutants." [89] The same Recommendation speaks of a desire to " improve the international acceptability of procedures for testing pollutants and contaminants " [90] and suggests ways this could be done. Subsections (a) and (b) of Recommendation 85 are even more to the point:

> It is recommended that any mechanism for co-ordinating and stimulating the actions of the different United Nations organs in connexion with environmental problems include among its functions:
> (a) Development of an internationally accepted procedure for the identification of pollutants of international significance and for the definition of the degree and scope of international concern;
> (b) Consideration of the appointment of appropriate inter-governmental expert bodies to assess quantitatively the exposures, risks, pathways, and sources of pollutants of international significance.[91]

This, of course, is just a beginning and it will be some time before one will be able to tell at least to what degree an international harmonisation of pollution standards is possible. Given different national and individual conceptions of how best to manage the environment and the various criteria which are still used to define " pollution," there is room for scepticism. More work needs to be done in defining clearly the types of criteria on which the experts should base their studies. General principles must be formed into tight policy guidelines which look ahead to both the problems to be faced and the economic and social goals sought. Once these standards, the " pollution limits," are established, then the negotiations must begin to attach the legal consequences to their infraction. That is a separate process. A general and meaningful concept of pollution will be established when the experts, working within the conceptual framework provided by the international community, are able to draw particular thresholds of legally-significant change that are accepted for those negotiations.

89 *Stockholm Declaration on the Human Environment*, Recommendation 74, pp. 40–41.
90 *Ibid.*
91 *Ibid.*, p. 84.